

Unit IV –INSTANT BASED LEARNING (K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning) from your Machine Learning syllabus, I have prepared **40 analytical assignment questions** aligned with relevant **UN Sustainable Development Goals (SDGs)** wherever applicable. The assignments emphasize data analysis, critical thinking, and real-world problem solving, matching the course outcomes.

Rubrics (Applicable to all Assignments)

Criterion	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Poor (1)
Problem Understanding	Clearly identifies problem and objectives	Mostly clear	Adequate	Limited understanding	Incorrect understanding
Data Collection & Preparation	Appropriate, clean dataset with preprocessing	Minor preprocessing issues	Basic preprocessing	Poor preprocessing	No preprocessing
Analysis using NumPy/Pandas	Correct and efficient analysis	Mostly correct	Basic analysis	Limited analysis	Incorrect analysis
Visualization & Interpretation	Appropriate graphs with meaningful insights	Good visualization	Basic charts	Minimal interpretation	No visualization
Innovation & SDG Mapping	Excellent real-world relevance with SDG linkage	Good linkage	Moderate linkage	Weak linkage	No SDG relevance
Documentation & Conclusion	Professional report with justified conclusions	Good report	Adequate report	Poor report	Incomplete

Total = 30 Marks

Unit IV Analytical Assignments with SDGs

No.	Assignment Question	SDG Alignment
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13	Apply KNN to classify marine pollution levels using ocean monitoring datasets.	SDG 14 – Life Below Water
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13. Apply KNN to Classify Marine Pollution Levels Using Ocean Monitoring Datasets

Objective: Use the K-Nearest Neighbors (KNN) algorithm to classify marine pollution levels, such as **Low, Medium, and High**, using ocean monitoring data.

Input Data: Ocean monitoring datasets containing features such as water temperature, pH, dissolved oxygen, turbidity, salinity, nitrate concentration, phosphate concentration, plastic concentration, and other pollution indicators.

Method:

1. Collect ocean monitoring data from different locations and time periods.
2. Clean the dataset by removing duplicate records and handling missing values.
3. Select relevant environmental and pollution-related features.
4. Normalize the numerical features because KNN uses distance calculations.
5. Divide the dataset into training and testing sets.
6. Select a suitable value of k , such as 3, 5, or 7.
7. For a new ocean sample, calculate its distance from all training samples.
8. Identify the nearest samples.
9. Count the pollution class of the nearest samples.
10. Assign the class having the highest number of votes.
11. Evaluate the classifier using accuracy, precision, recall, and confusion matrix.

KNN Distance: Euclidean distance can be calculated as:

where x is the new ocean observation and x_i is a training observation.

Pseudocode:

Algorithm Marine_Pollution_KNN

Input:

Ocean monitoring dataset D

Output:

Pollution level: Low, Medium, or High

1. Load ocean monitoring dataset
2. Remove duplicate and incorrect records
3. Handle missing values
4. Select relevant pollution features
5. Normalize all numerical features
6. Divide dataset into training and testing sets
7. Select value of K
8. For each test sample:
 - Calculate distance from every training sample
 - Select K nearest samples
 - Count class labels of K neighbors
 - Assign majority class to test sample
9. Calculate accuracy, precision, recall, and F1-score

10. Display predicted marine pollution level

End

Example Output:

Sample 1 → Low Pollution

Sample 2 → Medium Pollution

Sample 3 → High Pollution

Advantages: KNN is simple, easy to implement, and does not require a complex training process. It can classify pollution levels based on similarities between new observations and previously labeled ocean samples.

Limitations: KNN can become computationally expensive for very large datasets because it calculates distances to many training samples. Its performance is also sensitive to the choice of and feature scaling.

SDG 14 – Life Below Water

SDG 14 aims to **conserve and sustainably use the oceans, seas, and marine resources**. A KNN-based marine pollution classification system supports SDG 14 by helping identify areas with different pollution levels.

Application: Ocean monitoring stations can collect water-quality measurements, and the KNN model can classify each location as having low, medium, or high pollution. Authorities can use these classifications to identify polluted areas, monitor changes over time, and take appropriate environmental protection measures.

SDG Contribution: This system supports marine pollution monitoring, protection of aquatic ecosystems, sustainable use of marine resources, and data-driven environmental decision-making.

Expected Deliverables (Same as Unit V)

- Problem statement and relevant SDG
- Dataset description (Kaggle, UCI ML Repository, Data.gov.in, WHO, World Bank, etc.)
- Data preprocessing
- Model development using **KNN, LWR, RBF, or Case-Based Learning**
- Model evaluation (Accuracy, Precision, Recall, F1-score, RMSE, MAE, etc., wherever applicable)
- Result visualization
- Interpretation of results
- SDG-based recommendations
- Conclusion and references

These assignments are designed to satisfy the **Unit IV learning outcomes** on instance-based learning approaches including k-nearest neighbors, locally weighted regression, and case based reasoning while integrating the **UN Sustainable Development Goals (SDGs)** into practical machine learning data analysis tasks.